

Step 1: Title Page

- KhetHub Internship Program 2026
- **Task 4** – Smart Agriculture product proposal challenges
- Name:- Pranshu Tyagi
- Role: Web Development Intern
- Date:- 06/07/2026

This task is evaluating your **product thinking + web development knowledge**, not just coding. Since you're applying as a **Web Development Intern**, you should explain the full architecture in a professional way.

KhetHub Internship Task 4 Proposal

Smart Water Management & Irrigation Advisor

1. Problem Identification

Problem

Many farmers do not know **when** and **how much** to irrigate their crops.

As a result:

- 💧 Water is wasted.
- 🌿 Crops receive too much or too little water.
- 📉 Crop yield decreases.
- 💰 Electricity and diesel costs increase.

Research

According to agricultural studies, irrigation decisions are often based on experience rather than real-time field conditions. This leads to inefficient water usage, especially during uncertain weather.

2. Proposed Solution

Product Name

Smart Irrigation Advisor

Feature Overview

A digital feature inside KhetHub that recommends the best irrigation schedule using:

- Weather forecast
- Soil moisture
- Crop type

- Crop growth stage
 - Farm location
-

Target Users

- Small farmers
 - Medium farmers
 - Large farms
 - Agricultural advisors
-

Key Functionalities

Weather Prediction

Shows rainfall prediction.

Irrigation Recommendation

Example:

Your wheat field requires irrigation tomorrow morning for 35 minutes.

Crop Selection

Farmer selects:

- Wheat
 - Rice
 - Potato
 - Tomato
 - etc.
-

GPS Location

Automatically detects village/location.

Notification

Send reminder:

Irrigate your field at 6:00 AM tomorrow.

Water Usage History

Displays:

- Total irrigation
 - Water saved
 - Previous recommendations
-

Benefits for Farmers

- ✓ Save Water
 - ✓ Higher Yield
 - ✓ Lower Electricity Cost
 - ✓ Better Crop Health
 - ✓ Scientific Decision Making
-

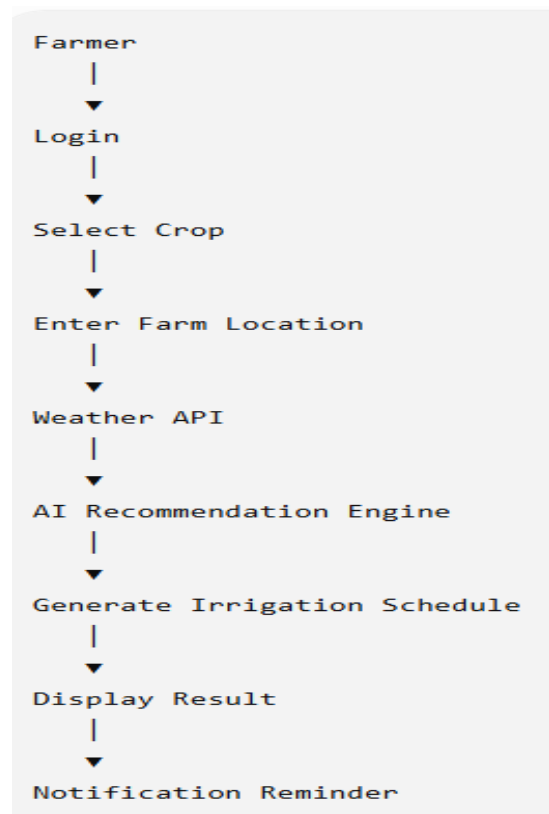
Why KhethHub Should Build It

This feature:

- Solves a real farming problem
 - Can reach millions of farmers
 - Improves customer engagement
 - Supports sustainable farming
 - Can later integrate with IoT devices
-



3. Product Workflow



Simple Wireframe

Smart Irrigation Advisor

Crop:
[Wheat ▼]

Location:
Meerut

Soil Moisture:
38%

Weather:
Rain Tomorrow

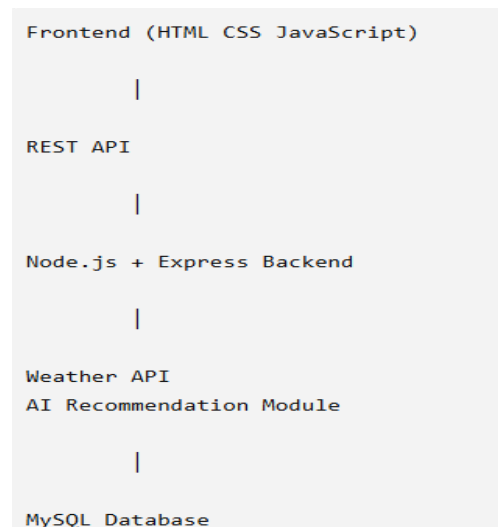
Recommendation

Irrigate Tomorrow
Duration:
35 Minutes

[Start Irrigation]

4. Technical Approach

System Architecture



Frontend

Technology:

- HTML
- CSS
- JavaScript

Pages:

- Login
- Dashboard
- Crop Selection
- Recommendation Screen
- History Page

Backend

Technology:

- Node.js
- Express.js

Responsibilities:

- User Authentication
 - Fetch Weather
 - Generate Recommendation
 - Store Farmer Data
 - Notification API
-

Database

Users Table

Field	Type
user_id	INT
name	VARCHAR
mobile	VARCHAR
village	VARCHAR

Farms

Field	Type
farm_id	INT
user_id	INT
crop	VARCHAR
area	FLOAT

Irrigation Records

Field	Type
record_id	INT
farm_id	INT
moisture	FLOAT
recommendation	TEXT
date	DATE

API Design

Login

```
POST /api/login
```

Request

```
</> JSON
{
  "mobile": "9876543210",
  "password": "123456"
}
```

Response

```
</> JSON
{
  "status": "success",
  "token": "JWT_TOKEN"
}
```

Get Recommendation

```
GET /api/recommendation
```

Response

«» JSON

```
{
  "crop": "Wheat",
  "soilMoisture": "38%",
  "weather": "Rain Tomorrow",
  "recommendation": "Irrigate Tomorrow Morning",
  "duration": "35 Minutes"
}
```

Save Recommendation

POST /api/history

Fetch History

GET /api/history

Backend Workflow

User Login

↓

Authentication

↓

Fetch Farm Details

↓

Call Weather API

↓

Analyze Soil Moisture

↓

Generate Recommendation

↓

Store Database

↓

Return JSON Response

↓

Display Dashboard

Bonus Features

AI Integration

AI predicts:

- Disease risk
 - Water requirement
 - Crop stress
-

Voice Assistant

Farmer asks:

"Kal paani dena hai?"

Assistant replies in Hindi.

Multilingual Support

- Hindi
 - English
 - Punjabi
 - Marathi
 - Gujarati
 - Tamil
-

IoT Integration

Connect:

- Soil Moisture Sensor
 - Smart Irrigation Pump
-

Offline Mode

Farmer can view the last downloaded recommendation without internet.

GPS Service

Automatically identifies:

- District
- Weather

- Nearby mandi

Predictive Analytics

Shows:

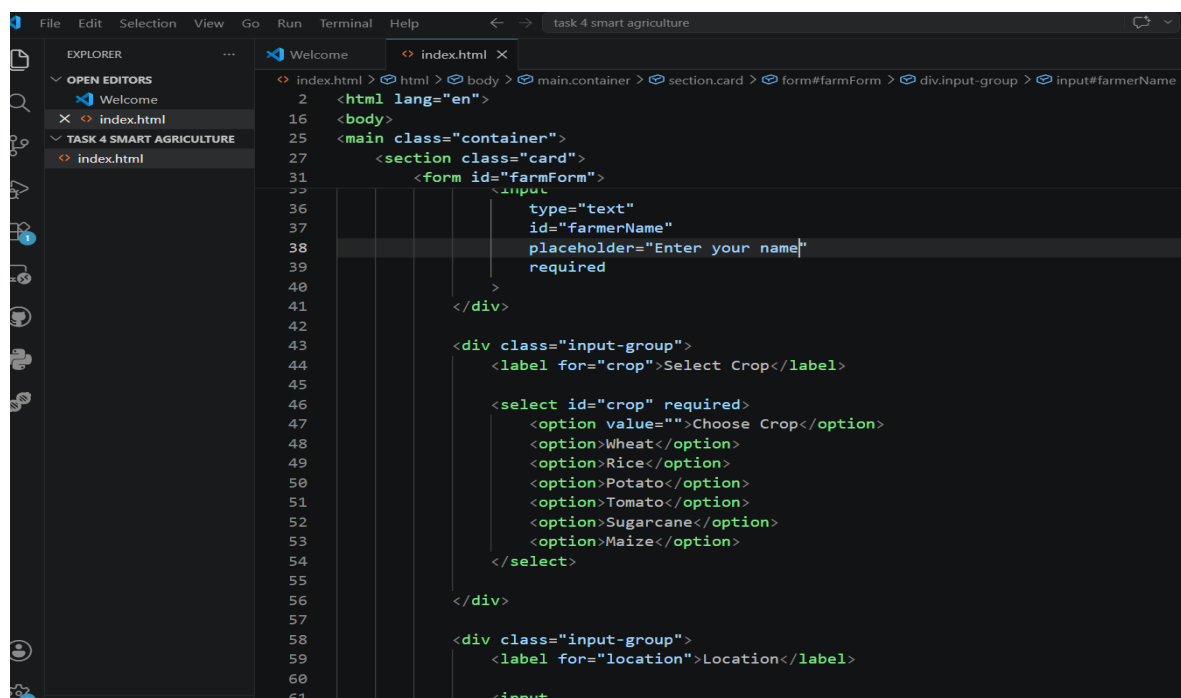
- Expected water requirement
- Rainfall prediction
- Water-saving statistics

UI Suggestion

Dashboard Sections

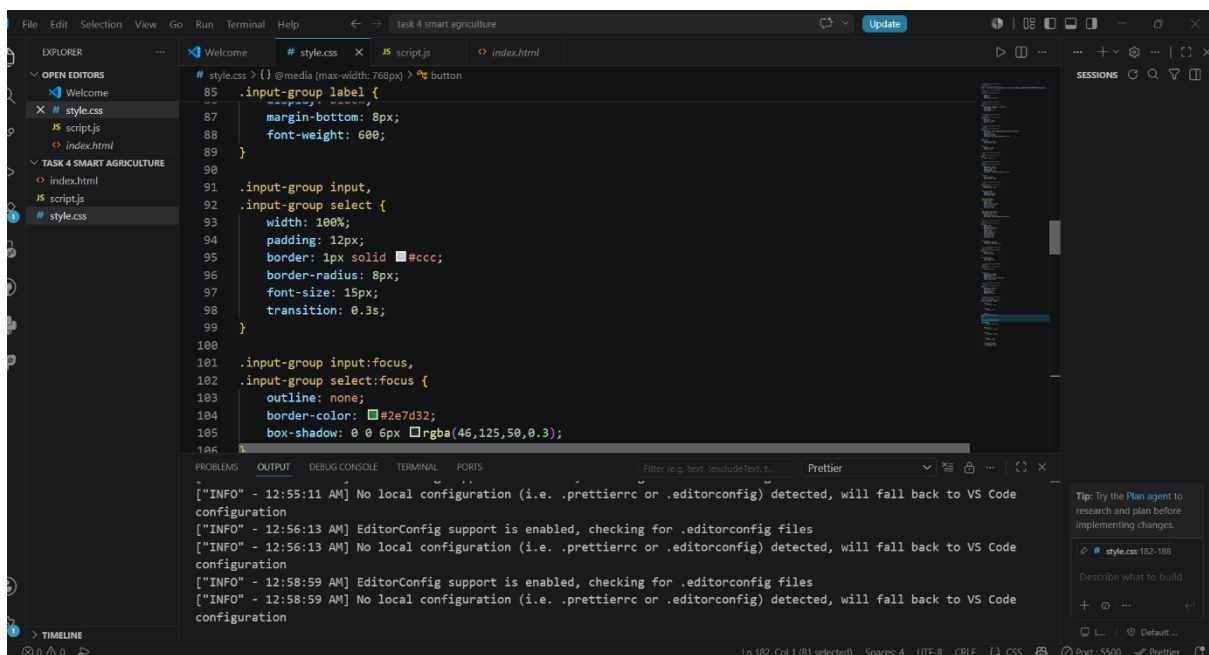
1.  Today's Weather
2.  Crop Details
3.  Irrigation Recommendation
4.  Water Usage Report
5.  Notifications
6.  History

Index.html

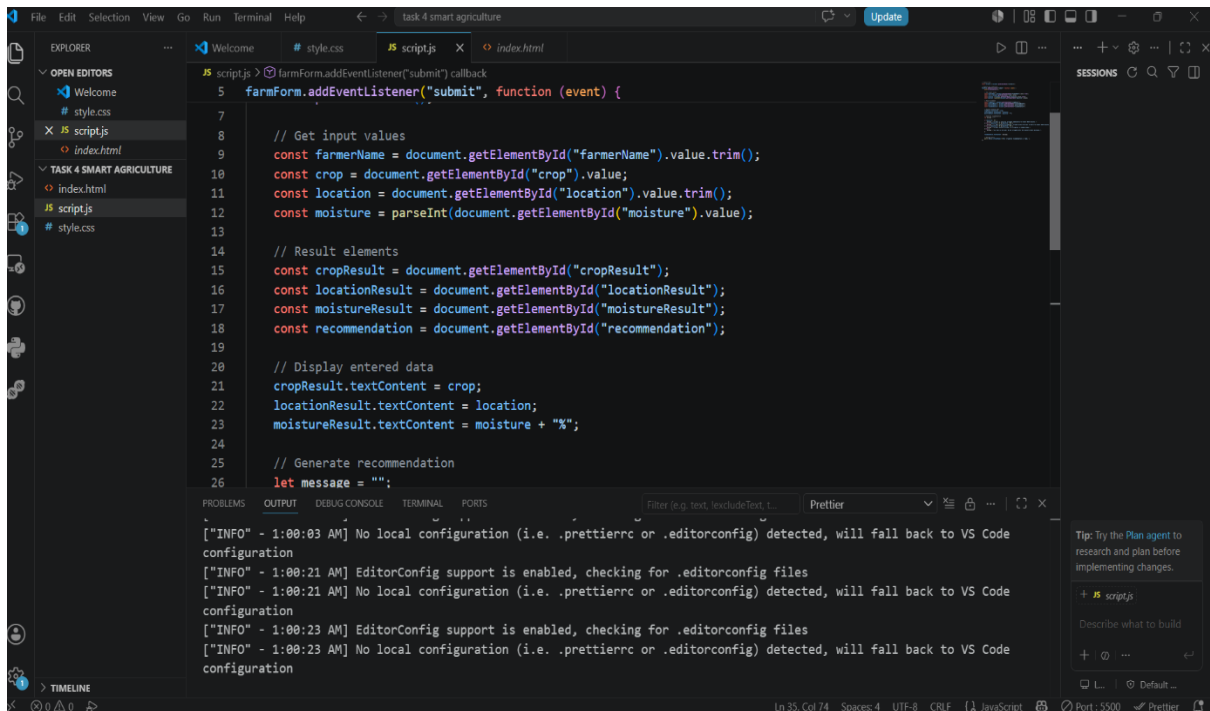


```
2 <html lang="en">
16 <body>
25 <main class="container">
27 <section class="card">
31 <form id="farmForm">
33 <input
36   type="text"
37   id="farmerName"
38   placeholder="Enter your name"
39   required
40 >
41 </div>
42
43 <div class="input-group">
44   <label for="crop">Select Crop</label>
45
46   <select id="crop" required>
47     <option value="">Choose Crop</option>
48     <option>Wheat</option>
49     <option>Rice</option>
50     <option>Potato</option>
51     <option>Tomato</option>
52     <option>Sugarcane</option>
53     <option>Maize</option>
54   </select>
55 </div>
56
57 <div class="input-group">
58   <label for="location">Location</label>
59
60   <input
```

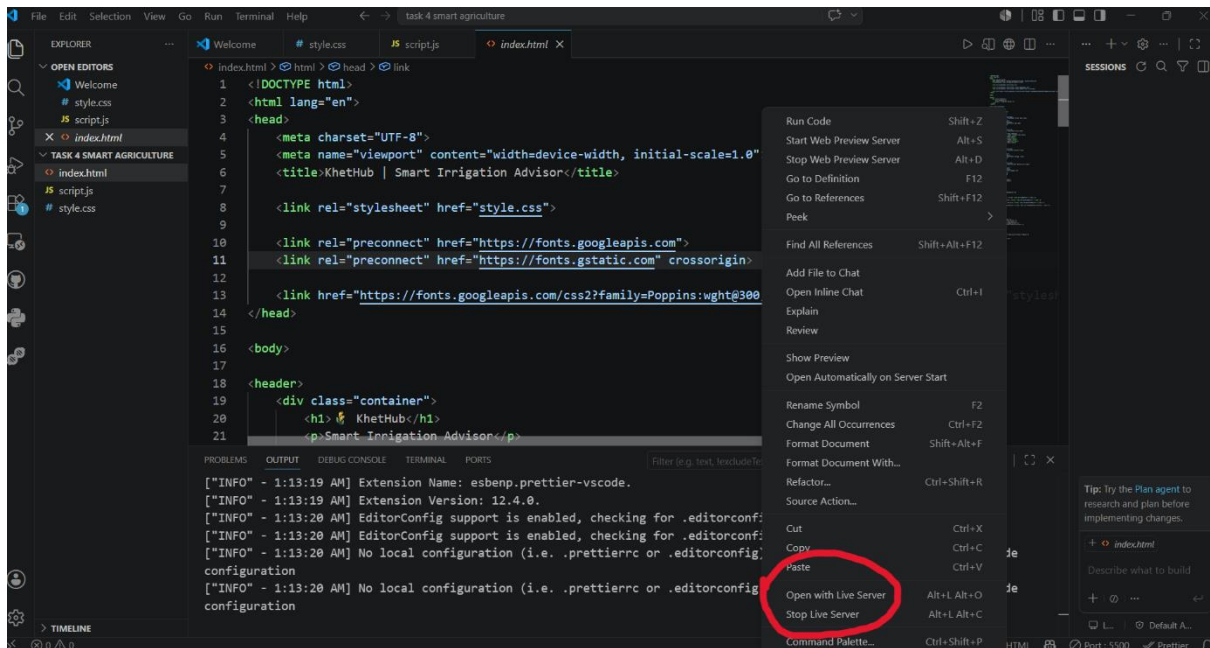
Style.css



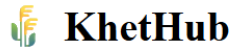
Script.js



Live server through index.html



Before



Smart Irrigation Advisor

Farm Details

Farmer Name

Select Crop

Location

Soil Moisture (%)

Irrigation Recommendation

Crop: --

Location: --

Soil Moisture: --

Recommendation: Waiting...

Benefits

-  Save Water
-  Improve Crop Health
-  Increase Crop Yield
-  Reduce Irrigation Cost
-  Smart Weather-Based Decisions

And after that

Smart Agriculture dashboard challenges

New tabXKhetHub | Smart Irrigation AdvisorX+

127.0.0.1:5500/index.html

Verify it's you...Chat

 **KhetHub**
Smart Irrigation Advisor

Farm Details

Farmer Name
Pranshu Tyagi

Select Crop
Rice

Location
meerut

Soil Moisture (%)
10

Get Recommendation

New tabXKhetHub | Smart Irrigation AdvisorX+

127.0.0.1:5500/index.html

Verify it's you...Chat

Irrigation Recommendation

Crop: Rice
Location: meerut
Soil Moisture: 10%
Recommendation: 🚰 Soil is very dry. Irrigate immediately for about 40–45 minutes.

Benefits

- 💧 Save Water
- 🌱 Improve Crop Health
- 📈 Increase Crop Yield
- 💰 Reduce Irrigation Cost
- 🌤️ Smart Weather-Based Decisions

